K667 — Pre-adjudication for THE BUILD: the thermal-odd-harmonic model gives a specific neutrino Δm^2-ratio TARGET (e^{4c}+1), and a precision flag on "odd cohomology vs odd-index Chern." A lead for the build, NOT a bank — the slope c is a fit until the geometry forces it.

Keeper | 2026-07-09 | Auditor pre-work while Lyra+Grace run the build. Two deliverables: (1) a concrete checkable target for the thermal τ-gradient, with the target-innocence discipline explicit; (2) a convention precision flag for Elie.

(1) The thermal-odd-harmonic model makes a SPECIFIC prediction — hand it to the build as the target

Model: three generations = odd harmonics h ∈ {1, 3, 5} of the stopped-pipe/quarter-wave resonator; mass thermally activated along the τ-gradient, m_i ∝ e^{c·h_i} (hot/extended end = heavier ⇒ + exponent, sign correct: gen-3 = mode 5 = heaviest). Oscillation energy = m^2.

Structural consequence (given the model, no fit yet): - $m_1{}^2$:$m_2{}^2$:$m_3{}^2$ ∝ e^{2c} : e^{6c} : e^{10c} - $\Delta m^2{}_{21}$ ∝ $e^{\{2c\}(e}${4c}−1); $\Delta m^2{}_{31}$ ∝ $e^{\{2c\}(e}${8c}−1) - $\Delta m^2{}_{31}/\Delta m^2{}_{21}$ = ($e^{\{8c\}-1)/(e}${4c}−1) = e^{4c} + 1 ← the ratio is forced to this form by the geometric series.

The observed ratio is 33. So the model lands iff e^{4c} = 32 = 2^{n_C}, i.e. the τ-gradient slope c = (n_C·ln2)/4 ≈ 0.866. Then: - $\Delta m^2{}_{31}/\Delta m^2{}_{21}$ = 2^{n_C} + 1 = 33 — a NEW form for the banked 33, alongside the Chern form N_c·c_2(Q^5) = 3·11. - The "+1" is AUTOMATIC (the +1 in x+1 from the geometric series) — this may be the mechanical origin of the corpus "+1 anomaly" (33=32+1, Λ exponent 281=280+1). Lead. - Falsifiable side-prediction: $m_1{}^2 = m_2{}^2$/(2^{n_C}−1) ⇒ lightest neutrino $m_1 \approx m_2/\sqrt{31}$ ≈ 1.5 meV, NOT zero (vs the seed's m_ν1=0). Distinguishes the thermal model from the massless-ground reading; within current bounds.

THE DISCIPLINE (loud) — this is a LEAD, not a bank

I fit c to hit 33. The clean values ($e^{\{4c\}=2}${n_C}, ratios 1:32:1024, 33=2^{n_C}+1, m_1≈1.5 meV) are seductive PRECISELY because I chose c. Target-innocence lens: the integers here (n_C, the +1) became clean only *after* I solved for the target ⇒ FIT-suspect until the geometry independently forces c = (n_C·ln2)/4. - The build's job: does the *geometric* τ-gradient (the substrate's own inverse-time profile) come out at slope (n_C·ln2)/4? If YES → 33 = 2^{n_C}+1 is DERIVED + m_1≈1.5 meV predicted → genuine bank. If the slope is anything else → I fit it, discard the clean story. - What the build gets from this: a *specific number to check the gradient against* (c ≈ 0.866 = (n_C ln2)/4), a ratio formula (e^{4c}+1), and a falsifiable m_1. That's the value of the pre-work — it turns "find the gradient" into "check the gradient against 0.866."

(2) PRECISION FLAG for Elie — "odd cohomology" vs "odd-index Chern class"

Elie's bridge: generations = "odd Chern classes {c_1,c_3,c_5} in odd cohomology {h^1,h^3,h^5} of Q^5" (T1929). Two different things: c_1,c_3,c_5 are odd-*index* Chern classes, but they live in *even* cohomology degrees (H^2, H^6, H^{10}) — a smooth complex quadric has cohomology only in even degrees. "Odd cohomology h^1,h^3,h^5" (odd *degree*) would be a different (and for a complex quadric, vanishing) object. - The resonator analogy (stopped pipe = odd *harmonics* {1,3,5}) matches the odd INDEX cleanly — c_1,c_3,c_5 are the 1st/3rd/5th classes, odd-indexed, three of them, capped at 5 because dim=n_C=5. That's the real, tight statement. - Ask Elie to pin T1929 to the primary source: is it odd-index classes (even cohomology) or genuinely odd-degree cohomology? The convergence is strong either way, but the phrase should be exact before the build reads the modes off it (pin-conventions-to-primary-sources discipline; genus-name-flipped-3× precedent).

Net

The thermal-odd-harmonic model is now a *checkable* model, not just a picture: it forces Δm^2ratio = e^{4c}+1, it hits 33 at the clean slope c=(n_C ln2)/4 (⇒ 33=2^{n_C}+1), and it predicts $m_1 \approx 1.5$ meV — ALL contingent on the geometry forcing that slope, which is the build's gate. And "odd Chern" needs pinning to index-vs-degree. Both delivered to the team as targets/flags, banked nothing.

— Keeper K667, 2026-07-09. Thermal-odd-harmonic model → $\Delta m^2{}_{31}/\Delta m^2{}_{21}$ = e^{4c}+1 (geometric-series-forced); = 33 = 2^{n_C}+1 at slope c=(n_C·ln2)/4 ≈ 0.866; predicts lightest ν $m_1 \approx 1.5$ meV (not 0). LEAD not bank — I fit c to 33; genuine only if the geometric τ-gradient independently forces that slope (the build's gate). Target-innocence discipline explicit. Precision flag: Elie's "odd cohomology $\{h^1, h^3, h^5\}$" vs odd-index Chern $\{c_1, c_3, c_5\}$ (even degrees) — pin T1929 to source; resonator matches odd-INDEX cleanly, capped at 3 by dim=n_C=5.